SAP-C02 Must-Know Definitions

The words the exam assumes you know · term and meaning only · core concepts first, then service-area terms

Frameworks & big ideas

Well-Architected: 6 pillars	**Operational excellence** (run, monitor, improve) · **Security** (protect data and systems) · **Reliability** (recover, meet demand) · **Performance efficiency** (right resources, used well) · **Cost optimization** (no waste) · **Sustainability** (less energy and resources)
Shared responsibility model	AWS secures *the cloud* (hardware, facilities, managed service); you secure *what you put in it* (data, IAM, configuration, OS on EC2). More managed = less for you.
Cloud Adoption Framework (CAF)	6 perspectives of organizational readiness: Business, People, Governance, Platform, Security, Operations.
Landing zone	Ready-made multi-account setup: OUs, guardrails, log archive and audit accounts, account vending (Control Tower).
Region / AZ / Local Zone / edge location	Geographic area / isolated data center group inside a Region / small AWS site near a city for low latency / CloudFront and Global Accelerator point of presence.
Managed vs serverless service	AWS runs the servers (you choose size) vs no servers to size at all; pay per use, scales to zero.
Control plane vs data plane	APIs that create or change resources vs the part that serves traffic. The data plane is more available, so failover should not depend on control-plane calls.
Defense in depth	Several layers of security (edge, network, instance, app, data) so one failure doesn't expose everything.
Zero trust	Never trust by network location; verify identity and context on every request.

Migration

7 Rs	**Retire** (switch off) · **Retain** (keep on-premises) · **Rehost** (lift and shift, e.g. MGN) · **Relocate** (move the hypervisor platform, e.g. VMware Cloud on AWS) · **Repurchase** (swap for SaaS) · **Replatform** (small changes to use managed services, e.g. DB → RDS) · **Refactor** (re-architect cloud-native)
Migration phases	Assess (business case) → Mobilize (plan, landing zone, skills) → Migrate and modernize (in waves).
Wave	Group of servers or apps moved together, usually because they depend on each other.
TCO	Total cost of ownership: everything it costs to run, on-premises vs AWS.
Homogeneous / heterogeneous migration	Same database engine / different engine (needs schema conversion first).
CDC (change data capture)	Keeps copying changes after the full load until cutover; gives minimal downtime.
Cutover	The moment traffic switches from the old system to the new one.
Strangler fig pattern	Replace a monolith piece by piece behind a routing layer until the old system can be removed.
Data residency / sovereignty	Data must stay in a given country/Region / data is subject to that country's laws and control.

Identity & federation

Federation	Users sign in with an outside identity provider and receive temporary AWS credentials; no IAM users needed.
IdP / SP	Identity provider proves who you are (AD FS, Okta, Entra ID) / service provider trusts it (AWS).
SAML 2.0	XML standard for workforce sign-in → AssumeRoleWithSAML or IAM Identity Center.
OIDC (OpenID Connect)	Token standard for web/mobile apps and CI/CD (e.g. GitHub Actions) → AssumeRoleWithWebIdentity, Cognito.
SCIM	Standard that syncs users and groups automatically from the IdP.
STS	Security Token Service: issues temporary credentials when a role is assumed.
Permission set	Identity Center template that becomes an IAM role in each assigned account.
Identity-based vs resource-based policy	Attached to a user/role ("what can I do") vs attached to a resource ("who can use me"). Cross-account needs an allow on both sides.
Trust policy	Part of a role that says who may assume it.
Permissions boundary	Maximum permissions a user or role can ever have; grants nothing by itself.
SCP / RCP	Organization guardrails: SCPs limit your principals, RCPs limit access to your resources; neither grants anything.
Policy evaluation	Explicit deny wins → otherwise an allow is needed → otherwise implicit deny.
RBAC vs ABAC	Access by role vs access by matching tags (attributes).
Confused deputy / ExternalId	A third party tricked into using its access for someone else; an ExternalId in the trust policy prevents it.
Least privilege	Only the permissions needed, nothing more.

Resilience

RTO	Recovery time objective: how long you can be down.
RPO	Recovery point objective: how much data (in time) you can lose.
High availability / fault tolerance / DR	Stays up with brief interruption (multi-AZ) / no interruption at all (full redundancy) / recovery from a large disaster (another Region).
4 DR strategies	Backup and restore → pilot light → warm standby → multi-site active-active (cheaper and slower → costlier and faster).
Availability "nines"	99.9% ≈ 8.8 h down/year · 99.99% ≈ 53 min · 99.999% ≈ 5 min.
Horizontal vs vertical scaling	Add more instances vs make one instance bigger.
Scalability vs elasticity	Can grow to handle more load vs grows and shrinks automatically with demand.
Static stability	Keeps working during a failure without launching or changing anything (capacity already in place).
Blast radius	How much breaks when one thing fails.

Cell-based architecture	Workload split into independent copies (cells), each serving part of the users, so a failure hits only one cell.
Loose coupling	Components talk through queues or events, so one slow or failed part doesn't break the others.
Idempotent	Running the same request twice has the same effect as once; needed with retries.
Exponential backoff with jitter	Retry with growing, randomized waits to avoid overloading a service.
Circuit breaker	Stop calling a failing dependency for a while and fail fast instead of piling up requests.
Graceful degradation	Keep core functions working with reduced features when something fails.
Synchronous vs asynchronous replication	Write confirmed only after the copy is written (no loss, more latency) vs copied afterwards (fast, some loss possible).
Dead-letter queue (DLQ)	Holds messages that failed processing too many times, for later review.

Data

Strong vs eventual consistency	A read always sees the latest write vs it may briefly see an older value.
OLTP vs OLAP	Many small transactions (RDS, Aurora, DynamoDB) vs big analytical queries (Redshift, Athena).
Data lake	Raw data of all kinds stored centrally in S3 and queried in place.
Partition (shard) key	Value that spreads data across partitions; a poor key creates a "hot" partition.
Lazy loading vs write-through cache	Fill the cache on a miss vs update the cache on every write.
TTL	Time to live: when cached data or an item expires automatically.
Encryption at rest / in transit	Stored data encrypted (KMS) / data moving over the network encrypted (TLS).
Envelope encryption	A data key encrypts the data; a KMS key encrypts the data key.
KMS key types	AWS owned (never visible) · AWS managed (e.g. aws/s3; you can't edit or share it) · customer managed (you control policy, rotation, cross-account use).

Integration & deployment

Queue vs pub/sub	One message to one consumer that pulls (SQS) vs one message pushed to many subscribers (SNS).
Fan-out	One event copied to several independent consumers (SNS → several SQS queues, EventBridge rules).
Microservices	App split into small independent services, each deployed and scaled on its own.
All at once / rolling / rolling + extra batch / immutable	Replace everything (downtime) / in batches (less capacity) / extra batch first (full capacity) / new instances, swap when healthy (safest rollback).
Blue/green	Two full environments; switch traffic, switch back to roll back.
Canary / linear	Small % first then the rest / equal steps over time; automatic rollback on alarms.
IaC / drift / change set	Infrastructure as code (CloudFormation, CDK) / real resources no longer match the template / preview of a stack update.
Golden AMI / immutable infrastructure	Pre-built, patched image / servers are replaced, never changed in place.

Governance & cost

Management account / member account / OU	Payer and owner of the organization / every other account / folder of accounts that policies attach to.
Delegated administrator	Member account allowed to run a service (GuardDuty, Security Hub, Backup) for the whole organization.
Controls (guardrails)	Preventive (block: SCP) · detective (find: Config) · proactive (check before deploy: CloudFormation Hooks).
Consolidated billing	One bill for the organization; volume tiers and discounts are shared.
Showback vs chargeback	Showing teams their cost vs actually billing it to them.
CapEx vs OpEx	Buying hardware up front vs paying for use.
On-Demand / Savings Plans / RI / Spot	Pay per use, no commitment / 1–3 yr $/hour commitment, flexible / 1–3 yr commitment to a configuration / spare capacity, up to ~90% off, 2-minute interruption notice.
Rightsizing	Matching instance size to actual use.
Multi-tenancy: silo vs pool	Each customer gets dedicated resources vs customers share resources (separated by tenant ID).

Networking

CIDR	IP range notation (10.0.0.0/16); overlapping ranges can't be routed together.
Public vs private subnet	Route to an internet gateway vs no direct internet route (outbound only via NAT).
Transitive routing	A → B → C. Peering doesn't allow it; Transit Gateway does.
VIF types	Private (one VPC / VGW) · transit (Transit Gateway via DX gateway) · public (AWS public endpoints).
BGP	Routing protocol on Direct Connect and VPN; decides the preferred path.
ECMP	Spreads traffic over several equal VPN tunnels (Transit Gateway only).
Anycast	Same IP announced from many locations; users reach the closest (Global Accelerator).
Gateway vs interface endpoint	Free route-table entry for S3/DynamoDB vs paid private IP (ENI) for most services.
Stateful vs stateless firewall	Return traffic allowed automatically (security group) vs rules needed both ways (network ACL).

Compute & scaling

Instance purchase options	On-Demand · Savings Plans · Reserved Instances · Spot · Dedicated Instances (single-tenant hardware) · Dedicated Hosts (whole server, socket/core visibility for licences) · Capacity Reservations (guaranteed capacity in an AZ, no discount by itself).
Placement groups	Cluster (packed in one AZ, lowest latency) · Spread (separate hardware, max 7 per AZ) · Partition (groups of racks, for Kafka/HDFS/Cassandra).
Auto Scaling policies	Target tracking (keep a metric at a value) · step (add/remove by alarm size) · scheduled (known times) · predictive (learns daily/weekly patterns).

Health check grace period	Time a new instance gets before health checks can mark it unhealthy.
Lifecycle hook	Pauses an instance at launch or termination so a script can run (e.g. copy logs).
Warm pool	Pre-initialized stopped instances ready to join the group quickly.
Instance refresh	Replaces Auto Scaling instances in batches with a new launch template.
Cold start	Delay when Lambda creates a new execution environment.
Reserved vs provisioned concurrency	Caps or guarantees how many Lambda instances a function can use vs keeps environments pre-warmed (removes cold starts).
Fargate vs EC2 launch type	Containers with no servers to manage vs containers on instances you manage (OS/host control, GPUs, custom AMIs).

Storage & databases

Block / file / object storage	Disk for one instance (EBS) / shared file system (EFS, FSx) / objects by key over HTTP (S3).
S3 storage classes	Standard · Intelligent-Tiering (unknown access) · Standard-IA / One Zone-IA (infrequent) · Glacier Instant / Flexible / Deep Archive (archive, cheaper as retrieval gets slower).
S3 versioning / lifecycle	Keeps every version of an object / moves or deletes objects automatically by age.
S3 Object Lock	WORM protection: governance mode (users with permission can remove) vs compliance mode (nobody, not even root, until retention ends).
S3 replication: CRR / SRR / RTC / Batch Replication	Copy to another Region / same Region / with a 15-minute SLA / copy objects that existed before the rule.
Presigned URL	Time-limited URL that lets someone upload or download one object without AWS credentials.
S3 access point / Multi-Region Access Point	Named entry point with its own policy (and optional VPC-only access) / one global endpoint routing to buckets in several Regions.
EBS gp3 / io2	General-purpose SSD (IOPS set independently of size) / provisioned-IOPS SSD for the highest performance.
Instance store	Fast local disk that is lost when the instance stops or terminates.
RDS Multi-AZ instance vs Multi-AZ DB cluster	One standby that can't serve reads (failover ~1–2 min) vs two readable standbys (failover ~35 s).
Read replica	Asynchronous read-only copy; can be promoted to a standalone database.
Aurora storage / replicas / global database	6 copies across 3 AZs / up to 15 readers with fast failover / storage-level copy to other Regions (RPO ~1 s).
Switchover vs failover (Aurora global)	Planned move with no data loss vs unplanned move during an outage (recent writes may be lost).
DynamoDB on-demand vs provisioned	Pay per request, no planning vs set capacity (with auto scaling); cheaper for steady load.
GSI vs LSI	Global secondary index (different partition key, add any time) vs local secondary index (same partition key, created with the table).
DynamoDB Streams / global tables / DAX	Change log of item updates / multi-Region active-active replicas / in-memory cache (microseconds).
RDS Proxy	Pools and reuses database connections; faster failover for clients.

Networking & content delivery (more)

Internet gateway / NAT gateway / egress-only IGW	Two-way internet for public subnets / outbound-only IPv4 for private subnets / outbound-only IPv6.
Virtual private gateway / Transit Gateway / DX gateway	VPN or DX attachment for one VPC / regional hub connecting many VPCs and on-premises / global object linking a DX connection to VPCs or TGWs in many Regions.
VPC peering / VPC sharing	Private link between two VPCs (non-transitive) / other accounts launch into subnets of one central VPC (RAM).
Endpoint service	The provider side of PrivateLink: a service behind an NLB (or GWLB) that consumers reach through interface endpoints.
Load balancer types	ALB (HTTP/HTTPS, layer 7, path/host rules) · NLB (TCP/UDP, layer 4, static IPs) · GWLB (inline security appliances).
Sticky sessions	Load balancer sends a user to the same target each time (cookie).
Route 53 routing policies	Simple · weighted (split %) · latency (fastest Region) · failover (primary/secondary) · geolocation (by user's country) · geoproximity (by distance, with bias) · multivalue (several healthy IPs).
Alias record	Route 53 record pointing to an AWS resource (ALB, CloudFront, S3); works at the zone apex, no query charge.
Evaluate target health	Alias record checks the health of the resource it points to before answering.
CloudFront origin / behavior / OAC	Where content comes from / rule matching a path to an origin and cache settings / origin access control that keeps an S3 bucket private.
Origin group	Primary and secondary origin; CloudFront fails over automatically on errors.
Signed URL vs signed cookie	Access to one private file vs to many files.
CloudFront Functions vs Lambda@Edge	Tiny, very fast JavaScript at viewer events vs fuller functions (network access, request body) at viewer or origin events.
API Gateway types	REST (usage plans, API keys, caching, private APIs) · HTTP (cheaper, simpler, JWT auth) · WebSocket (two-way messages). Endpoints: edge-optimized, Regional, private.
Global Accelerator	Two static anycast IPs that route users over the AWS network to the nearest healthy Region (TCP/UDP).

Security services (one line each)

CloudTrail	Records API calls: who did what, when, from where.
AWS Config	Records resource configuration over time and checks it against rules.
GuardDuty	Threat detection from logs (CloudTrail, VPC Flow Logs, DNS): compromised credentials, crypto-mining.

Inspector	Scans EC2, container images and Lambda for software vulnerabilities.
Macie	Finds sensitive data (PII) in S3.
Detective	Investigates the root cause of a finding (graph of activity).
Security Hub (CSPM)	Collects findings in one place and checks best-practice standards.
IAM Access Analyzer	Finds resources shared outside your account or organization; generates least-privilege policies.
WAF / Shield / Firewall Manager	Filters web requests (layer 7) / DDoS protection (Advanced: 24/7 team, cost protection) / applies WAF and firewall rules across accounts.
Network Firewall	Managed stateful firewall for VPC traffic (filtering, IPS, domain lists).
KMS key rotation / multi-Region keys / grants	New key material automatically or on demand / same key copied to other Regions / temporary, programmatic permission to use a key.
CloudHSM	Single-tenant hardware security module that you control.
Secrets Manager vs Parameter Store	Secrets with built-in rotation and cross-account sharing vs configuration and simple secrets, no built-in rotation.

Monitoring & operations

Metrics / logs / traces	Numbers over time (CloudWatch) / event records (CloudWatch Logs) / path of a request across services (X-Ray).
CloudWatch alarm / composite alarm	Fires when a metric crosses a threshold / combines several alarms.
SLO / error budget	Target such as 99.9% of requests under 400 ms / how much failure is left before the target is missed.
Synthetic canary	Scripted test that runs on a schedule to check an app works, even with no real users.
Systems Manager pieces	Session Manager (shell without SSH) · Run Command · Patch Manager · State Manager (keep configuration) · Automation (runbooks).
EventBridge	Event bus that routes events by content rules; also schedules and archive/replay.
Step Functions Standard vs Express	Long (up to 1 year), exactly-once workflows vs short (up to 5 minutes), high-volume ones.
Fault Injection Service (FIS)	Runs controlled failure experiments with automatic stop conditions.
Game day	Planned practice of a failure scenario with the team.
Runbook vs playbook	Steps for a routine operation vs steps to investigate and respond to an incident.

Migration tools (one line each)

AWS Transform	AI-assisted discovery, planning and modernization (VMware, .NET, mainframe); replaces Migration Hub / Application Discovery Service for new customers.
Migration Evaluator	Builds a TCO business case from inventory and utilization data.
MGN (Application Migration Service)	Continuous block-level replication of whole servers; rehost with a short cutover.
DMS / SCT	Moves database data (full load + CDC) / converts schema and code to a new engine.
DataSync	Online, scheduled, verified copy of files/objects into S3, EFS or FSx.
Storage Gateway	On-premises access to cloud storage: S3 File Gateway (NFS/SMB), Volume Gateway (iSCSI), Tape Gateway (virtual tape).
Transfer Family	Managed SFTP/FTPS/FTP/AS2 endpoints into S3 or EFS.
Elastic Disaster Recovery	Continuous replication of servers to a low-cost staging area; launch on AWS in minutes, fail back later.
Snowball Edge / Data Transfer Terminal	Shipped device for offline transfer (existing customers only) / AWS facility where you plug in your own drives.